

Session 4: Data Retrieval with SELECT Trainer Coverage: SELECT queries, filtering, sorting, aliases. Aggregate functions (SUM, AVG, COUNT). Demo: Generate “total expenses” by category.

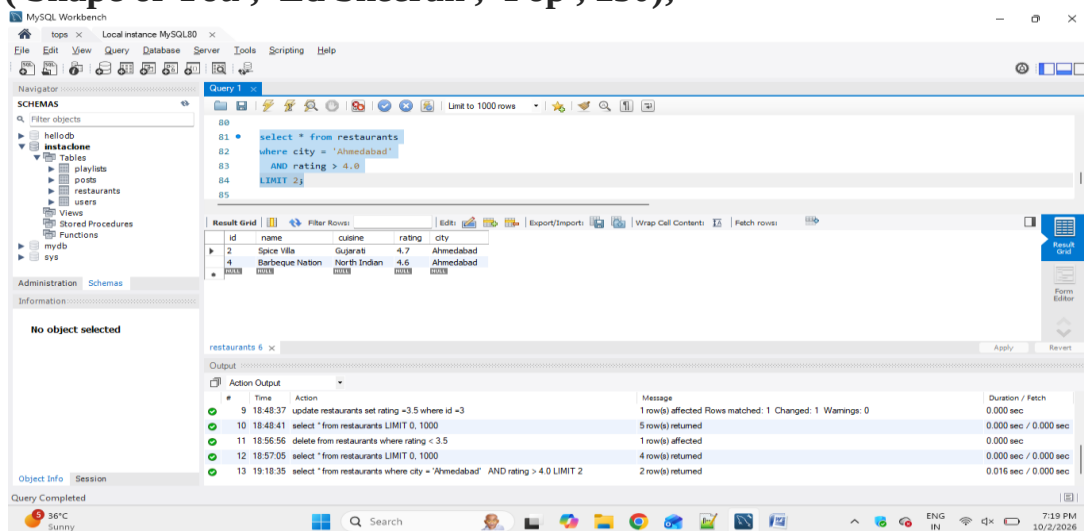
Q:-1. Create a table called Playlist with columns: id, song_name, artist, genre, and play_count. Insert at least 5 sample records representing your favorite songs from Spotify.

Create Table:-

```
create table playlist(  
id integer primary key auto_increment,  
song_name text,  
artist text,  
genre text,  
play_count integer);
```

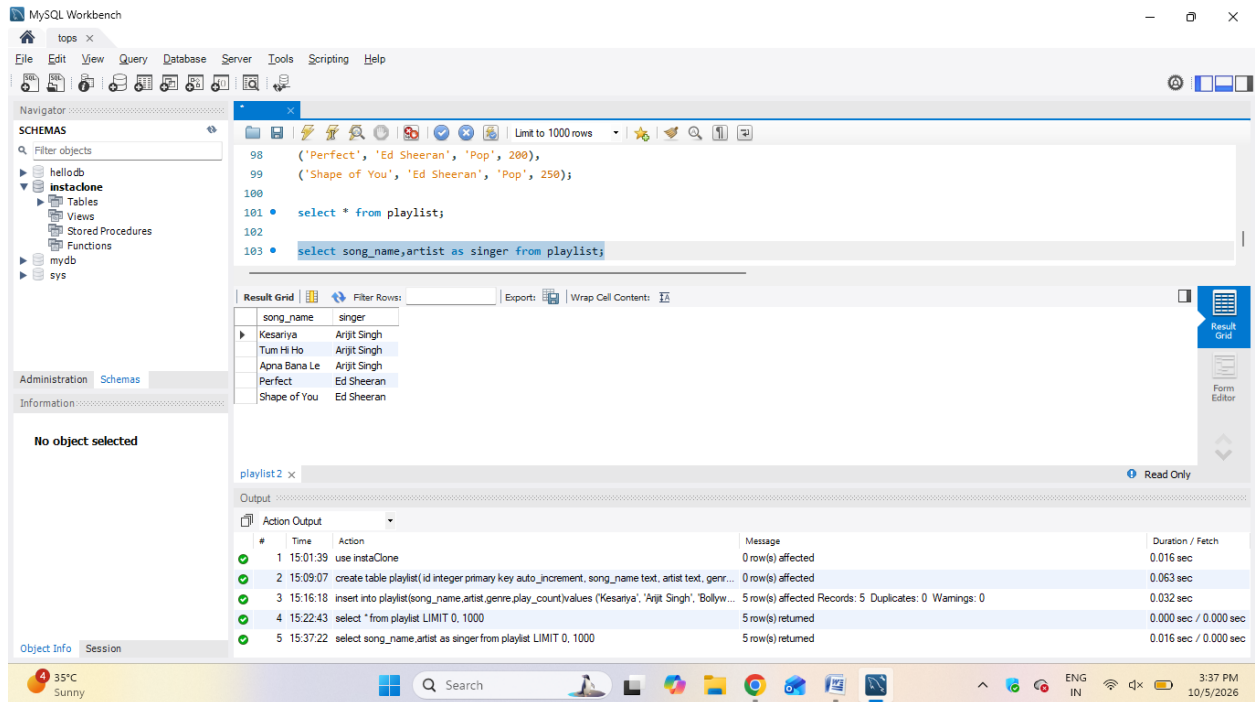
Insert Data:-

```
insert into playlist(song_name,artist,genre,play_count)values  
( 'Kesariya', 'Arijit Singh', 'Bollywood', 120),  
( 'Tum Hi Ho', 'Arijit Singh', 'Bollywood', 150),  
( 'Apna Bana Le', 'Arijit Singh', 'Romantic', 100),  
( 'Perfect', 'Ed Sheeran', 'Pop', 200),  
( 'Shape of You', 'Ed Sheeran', 'Pop', 250);
```



Q:-2. Write a SELECT query to display only the song_name and artist from the Playlist table, but show the artist column as 'Singer' using an alias.

select song_name,artist as singer from playlist;



Q:-3. Write a SELECT query to find all songs in the Playlist table where the genre is 'Pop' and play_count is greater than 100. Sort the results by play_count in descending order.

select *from playlist where genre ='Pop' and play_count>100 order by play_count desc;

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' panel with a tree view containing 'hellodb', 'instadone', 'Tables', 'Views', 'Stored Procedures', 'Functions', 'mydb', and 'sys'. The main editor window contains the following SQL queries:

```

100
101 • select * from playlist;
102
103 • select song_name,artist as singer from playlist;
104
105 • select *from playlist where genre = 'Pop' and play_count>100 order by play_count desc;

```

The 'Result Grid' shows the results of the first query (line 101):

	id	song_name	artist	genre	play_count
▶	5	Shape of You	Ed Sheeran	Pop	250
▶	4	Perfect	Ed Sheeran	Pop	200

The 'Output' panel shows the 'Action Output' for the executed queries:

#	Time	Action	Message	Duration / Fetch
2	15:09:07	create table playlist(id integer primary key auto_increment, song_name text, artist text, ge...	0 row(s) affected	0.063 sec
3	15:16:18	insert into playlist(song_name,artist,genre,play_count)values ('Kesariya', 'Arijt Singh', 'Boly...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.032 sec
4	15:22:43	select * from playlist LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
5	15:37:22	select song_name,artist as singer from playlist LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec
6	15:51:06	select *from playlist where genre = 'Pop' and play_count>100 order by play_count desc LI...	2 row(s) returned	0.000 sec / 0.000 sec

Q:-4. Use the COUNT aggregate function to find out how many songs in your Playlist table belong to the genre 'Hip-Hop'.

select count(*) as hip_hop_songs from playlist where genre ='Hip-Hop';

The screenshot shows the MySQL Workbench interface with the following SQL queries in the editor:

```

102
103 • select song_name,artist as singer from playlist;
104
105 • select *from playlist where genre = 'Pop' and play_count>100 order by play_count desc;
106
107 • select count(*) as hip_hop_songs from playlist where genre = 'Hip-Hop';

```

The 'Result Grid' shows the result of the third query (line 107):

	hip_hop_songs
▶	0

The 'Output' panel shows the 'Action Output' for the executed queries:

#	Time	Action	Message	Duration / Fetch
3	15:16:18	insert into playlist(song_name,artist,genre,play_count)values ('Kesariya', 'Arijt Singh', 'Boly...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.032 sec
4	15:22:43	select * from playlist LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
5	15:37:22	select song_name,artist as singer from playlist LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec
6	15:51:06	select *from playlist where genre = 'Pop' and play_count>100 order by play_count desc LI...	2 row(s) returned	0.000 sec / 0.000 sec
7	16:41:30	select count(*) as hip_hop_songs from playlist where genre = 'Hip-Hop' LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

**Q:-5. Write a SELECT query that shows the total number of plays (SUM of play_count) for each genre in your Playlist table, grouped by genre.

Hint: Use GROUP BY with the SUM function to aggregate play counts by genre.**

select genre,sum(play_count) as total_plays from playlist group by genre;

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'hellodb' selected. The main editor window contains the following SQL query:

```
104
105 • select *from playlist where genre ="Pop" and play_count>100 order by play_count desc;
106
107 • select count(*) as hip_hop_songs from playlist where genre ="Hip-Hop";
108
109 • select genre,sum(play_count) as total_plays from playlist group by genre;
```

The 'Result Grid' shows the output of the third query:

genre	total_plays
Bollywood	270
Romantic	100
Pop	450

The 'Output' tab at the bottom shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
4	15:22:43	select *from playlist LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
5	15:37:22	select song_name,artist as singer from playlist LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec
6	15:51:06	select *from playlist where genre ="Pop" and play_count>100 order by play_count desc	2 row(s) returned	0.000 sec / 0.000 sec
7	16:41:30	select count(*) as hip_hop_songs from playlist where genre ="Hip-Hop" LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
8	16:44:06	select genre,sum(play_count) as total_plays from playlist group by genre LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec